Digital Research Toolkit for Linguists

Week 2: Looking at data

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Psycholinguistics and Cognitive Modeling Lab

Questions?

Homework

Goal: Check if you can follow instructions & make data.

- Experiment: Some people took the experiment multiple times??
- Experiment: there were some curious answers, like “z” or “NOTHING”

Table of contents

Data → Understand → Communicate → Share

create, find, data types, formats, encoding

R, packages, import, clean, transform, debug, visualize, export

document, create clean and beautiful reports

connect, collaborate, backup, replicate

Data sources

Metadata

"Data about data". It describes information like when, where, and how something was created or used.

- Name
- (IP) address and location
- Age
- Gender
- Ethnicity
- Health data
- Spending habits
- Search history
- Date of creation/modification
- Document's size
- ...

Evil-ish (Meta)data Uses

Tracking & Profiling

A woman in her mid-20s in a relationship for 2 years → Wedding

Buying unscented lotion, magnesium supplements → Pregnancy

Purchased a shower curtain, visited OBI → Moved recenty

Deleting food delivery apps → Health/diet/fitness

Visits second-hand stores, cheap supermarkets → Loan sharks

Browsing job listings late at night → Target for recruiters

Evil-ish (Meta)data Uses

Privacy Invasion

Political views, religion, sexual orientation, health status.

Location Tracking

Monitor movements, targetting, stalking, survaillance, advertising.

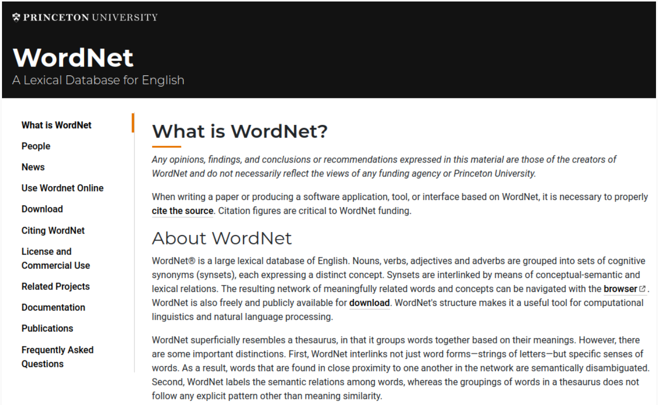
Social engineering, fraud, surveillance, government, competitive espionage, identity theft...

Corpus

Collection of written or spoken texts. Used to study language: how words and phrases are used, analyze content and semantic/syntactic/semiotic/... patterns, track language changes, language variation, cultural trends and societal issues, machine translation, train AI/LLM, lexicography & more.

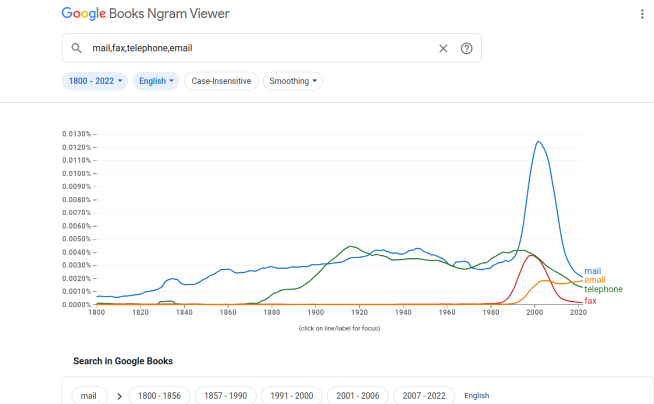
- → WordNet
- → Google Ngrams
- → Dependency Treebank
- → DeReKo
- → DECOW Corpora
- → Kaggle Twitter datasets

Humanities Data Sources: WordNet



A lexical database of English, grouping words into sets of synonyms, providing definitions and relationships (e.g. semantic analysis, NLP)

Humanities Data Sources: Google Ngrams



A tool that charts the frequencies of words/phrases in books over time (EN, DE, FR, CN, ...), based on the Google Books corpus (e.g. language change, cultural analysis).

Humanities Data Sources: Dependency Treebank

Universal Dependencies

Universal Dependencies (UD) is a framework for consistent annotation of grammar (parts of speech, morphological features, and syntactic dependencies) across different human languages. UD is an open community effort with over 600 contributors producing over 200 treebanks in over 150 languages (see the bottom of this page for updated numbers from the latest release). If you are new to UD, you should start by reading the first part of the Short Introduction and then browsing the annotation guidelines.

- Short introduction to UD
- UD annotation guidelines
- More information on UD:
  - How to contribute to UD
  - Tools for working with UD
  - Changes to the UD guidelines
  - UD-related events
  - Projects related to UD
- Query UD treebanks online:
  - PML Tree Query maintained by the Charles University in Prague
  - TEITOK maintained by the Charles University in Prague
  - Grew-match maintained by Inria in Nancy
  - INESS maintained by the University of Bergen
- Download UD treebanks

If you want to receive news about Universal Dependencies, you can subscribe to the UD mailing list. If you want to discuss individual annotation questions, use the Github issue tracker.

A parsed text corpus that represents syntactic structures using dependencies between words (e.g. syntactic analysis, machine translation).

Humanities Data Sources: Deutsches Referenzkorpus

Ausbau und Pflege der Korpora geschriebener Gegenwartssprache

Die weltweit größte Sammlung deutschsprachiger Korpora als empirische Basis für die linguistische Forschung

Das Deutsche Referenzkorpus – DeReKo

Die Korpora geschriebener Gegenwartssprache des IDS

- bilden mit 61,5 Milliarden Wörtern (Stand 31.01.2025) die weltweit größte linguistisch motivierte Sammlung elektronischer Korpora mit geschriebenen deutschsprachigen Texten aus der Gegenwart und der neueren Vergangenheit.
- sind über *COSMAS II* und *KorAP* kostenlos abfragbar
- enthalten belletristische, wissenschaftliche und populärwissenschaftliche Texte, eine große Zahl von Zeitungstexten sowie eine breite Palette weiterer Textarten und werden kontinuierlich weiterentwickelt.
- werden im Hinblick auf Umfang, Variabilität, Qualität und Aktualität akquiriert und erlauben in der Nutzungsphase über *COSMAS II* und v.a. *KorAP* die Komposition virtueller Korpora, die repräsentativ oder auf spezielle Aufgabenstellungen zugeschnitten sind.
- enthalten ausschließlich urheberrechtlich abgesichertes Material.
- Details zum DeReKo-Inhalt siehe *Archiv*

A vast collection of contemporary written German texts, accessible via the COSMAS II interface (e.g. corpus linguistics, lexicography).

Humanities Data Sources: Completion experiment

Q: Can a man marry his widow's sister? No 👻💀🪦

Q: According to the Bible, how many animals of each kind did Moses take on the ark? can't answer

Q: How many numbers are in the English alphabet? can't answer

Moses Illusions or *semantic illusions* occur when readers fail to recognize inconsistency in a text even if they were warned and know the correct word (Erickson and Mattson 1981).

Moses Illusion Experiment

Q: Can a man marry his widow's sister? No 👻💀🪦

Data that we got:

- Answer to the Moses illusion and the control questions
- Answer to the distractor questions
- Answer time
- ...

“Margaret Thatcher was the former president of which country?”

Expected: can’t answer

- I don’t know
- No idea
- UK
- United Kingdom
- can’t answer
- don’t know
- z

"Margaret Thatcher was the former prime minister of which country?"

Expected: UK or don't know

- Great Britain
- I don't know
- US
- United Kingdom
- britain
- uk

“What is the German inventor Johannes Gutenberg famous for?”

Expected: printing or don’t know → Instruction

- Book printing
- Don’t know
- First Printer
- Letterpress
- No idea
- Printer
- Printing
- Printing press
- don’t know
- inventing the typing machine
- mechanized printing
- movable type printing press
- printing press
- z

“Which Nordic country are coconut trees native to?”

Expected: can’t answer $\rightarrow$ Instruction

- Can’t answer
- No idea
- Sweden
- any Nordic country
- can’t answer
- can’t know
- coconut in the north?
- don’t know
- z

Data types: What the researcher sees

- response
- reaction times
- acceptability judgments
- completion
- recordings
- transcription, annotation
- brain waves
- texts
- video
- images
- …

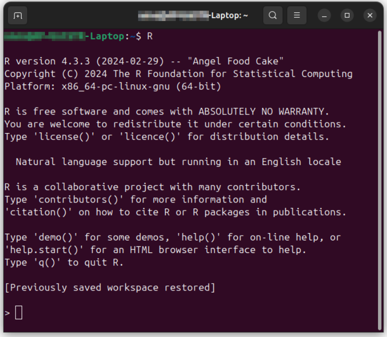
→ text files (`txt, csv, tsv`)

Data types: What the statistitian sees

nominal	marital status, religion
ordinal	grades, energy efficiency classes
interval	IQ, temperature in C and F
ratio	reaction times, population

R



R

A **language & environment** for statistical computing & graphics. It's used by scientists, data miners & statisticians for **data analysis & developing statistical software**.

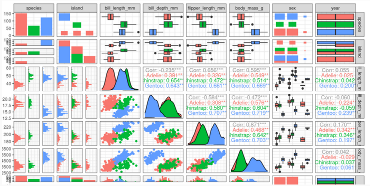
Base + packages

Developed by R Core Team & the R Foundation for Statistical Computing `https://www.r-project.org/`

Why botheR?

most popular in data science

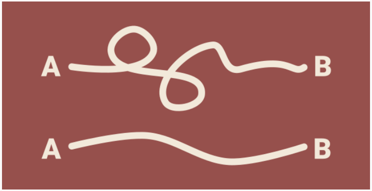
used in almost every industry



powerful visualizations

free, open source

Why botheR?



(relatively) easy to learn

gateway to lucrative careers

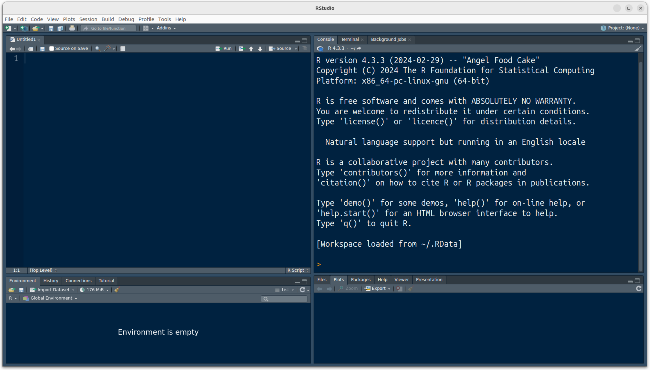
Bonus: you can develop cool web apps

`https://ceefluz.shinyapps.io/radar/`

`https://gallery.shinyapps.io/087-crandash/`

Working with RStudio

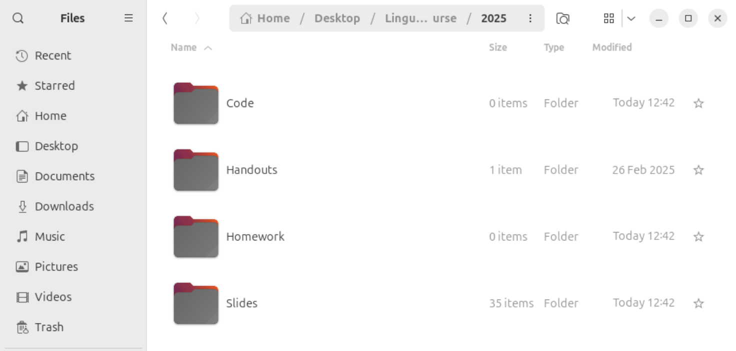
RStudio Integrated Development Environment (=IDE)



- free, open source, “friendlier” than pure R
- developed by Posit, who develop, maintain, and promote many awesome packages (e.g. tidyverse, RMarkdown, knitr) and Quarto
- source, console, environment, plots, packages, help

Make a home for your project

Create a new folder and set it as **the working directory**.

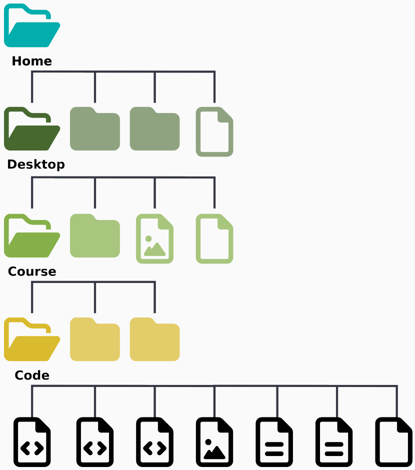


- where R will look for files
- where R will save visible and hidden files
- where R will automatically load files from

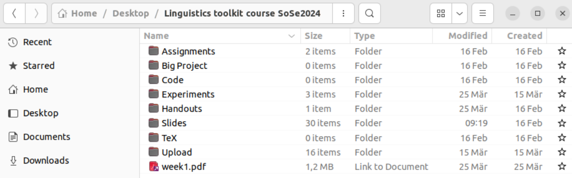
File structures and directories

A **directory** or **folder** is a container for storing files or other folders.

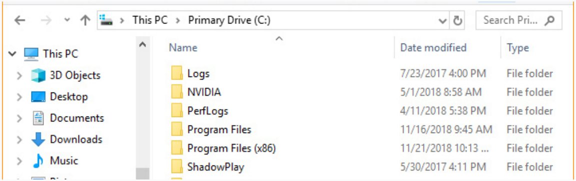
File structure or **file hierarchy** or **folder organization** is the way these containers are organized.



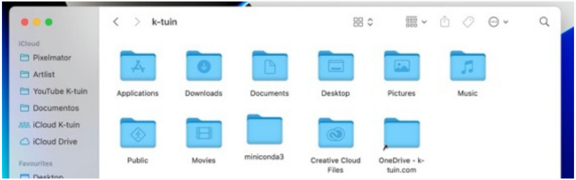
We’re different and we’re the same



Ubuntu

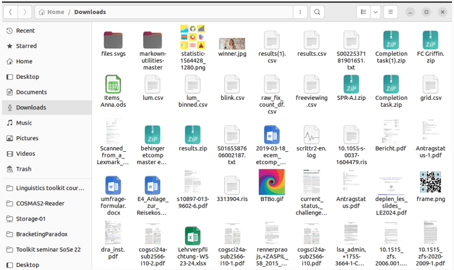


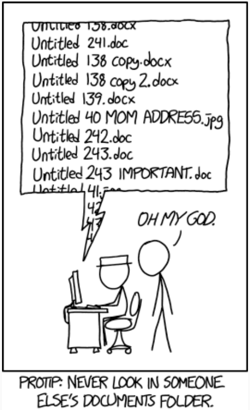
Windows



Mac

Why it matters





Don't do this to yourself, your future self, and your collaborators.

Consistent structure

Save space & time, establish expectations & be consistent, avoid confusion & repetition → efficient, transparent, and scalable project.

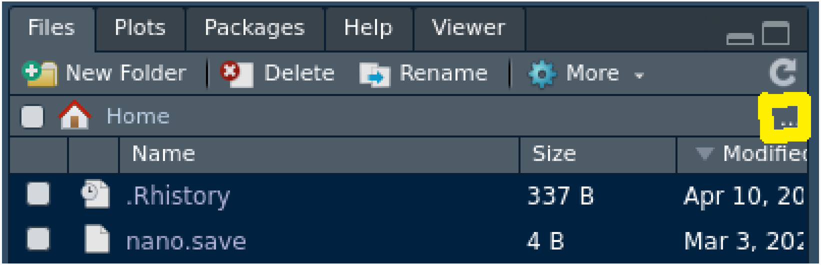
Meaningful names

Keep directory names short and descriptive.

```
📁 project_name/
├── 📁 data/              # Raw and processed data
│   ├── raw/              # Original, unmodified data
│   ├── processed/        # Cleaned, transformed, or filtered data
│   ├── metadata/         # Data descriptions, schemas, or notes
├── 📁 scripts/           # Code for analysis, preprocessing, and visualization
├── 📁 results/           # Generated outputs (tables, figures, models)
├── 📁 reports/           # Papers, presentations, or documentation
├── 📁 references/        # Papers, citations, and related documents
├── 📁 notebooks/         # Jupyter/RMarkdown/Quarto notebooks
├── 📜 README.md          # Overview of the project
```

Generated using OpenAI's ChatGPT on March 5, 2025

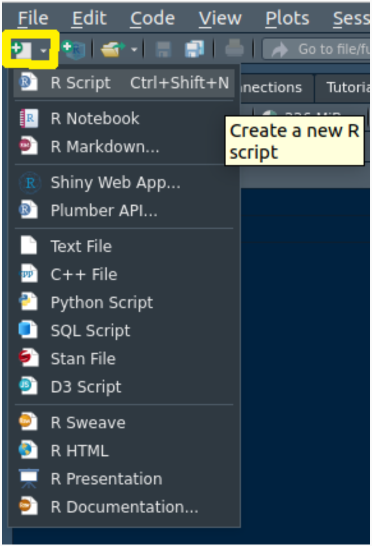
Make a home for your project



`setwd("PATH")` set working directory

`getwd()` check working directory

Don’t You Forget About Me



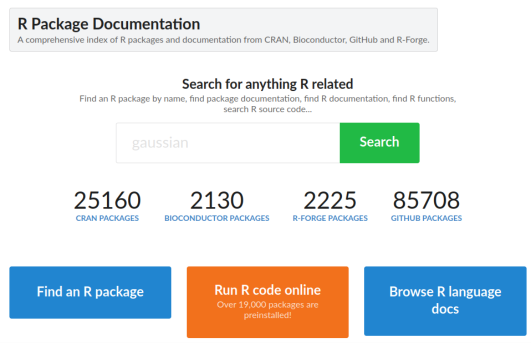
Create a new script for this class and save it in your **working directory**.

R packages

Packages

are collections of functions and/or datasets with a common theme (e.g. statistics, spatial analysis, plotting).

Most packages are available through the **Comprehensive R Archive Network (CRAN)** and on **GitHub**.

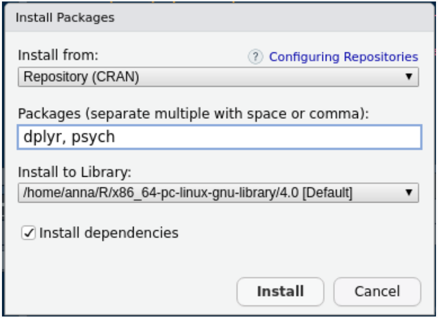


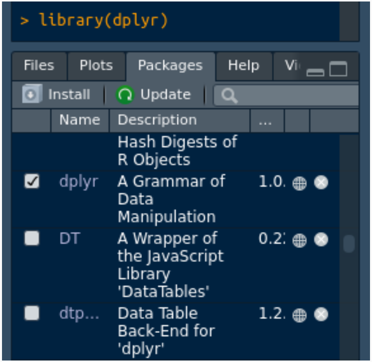
Installing, loading, checking

`install.packages(c("NAME","NAME2"))` install from repository

`library(NAME)` load into workspace (=activate)

`sessionInfo()` collect information about current R session





Questions?

Wrap-up

Summary

- Metadata
- Data sources
- Data types
- R and RStudio IDE
- Packages (installing and loading)
- (Working) directories and file hierarchy
- **Next week:** Looking at (your) data and R programming basics
- **Want to read more?** QCBS R Workshop Series

Homework assignment due April 21st 12:00

- Read chapters 2, 6 and 7 of *R for Data Science* (Wickham et al. 2023)
- Complete assignment 2 ($\rightarrow$ ILIAS)
- Name your files with name and assignment number, e.g. `pryslopskaassignment2.R` and `pryslopskaassignment2.png`

References

Erickson, Thomas D and Mark E Mattson (1981). **“From words to meaning: A semantic illusion”.** In: *Journal of Verbal Learning and Verbal Behavior* 20.5, pp. 540–551. DOI: 10.1016/s0022-5371(81)90165-1.

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund (2023). ***R for data science: import, tidy, transform, visualize, and model data.*** 2nd ed. O’Reilly Media, Inc. URL: https://r4ds.hadley.nz/.